

COVID-19: Out of Hospital Cardiac Arrest

Section 1: Case Summary

Scenario Title:	COVID-19: Out of Hospital Cardiac Arrest
Keywords:	COVID-19, cardiac arrest, EHS, resuscitation
Brief Description of Case:	The patient is found by a friend unresponsive after a 7 day history of cough and shortness of breath. He immediately receives bystander CPR. An advanced care paramedic crew attends the scene and manages a ventricular fibrillation arrest prior to transporting to hospital. The patient goes into cardiac arrest again shortly before arriving in the emergency department. The team will need to prepare for the patient's arrival and then manage a cardiac arrest using appropriate precautions for suspected COVID-19.

Goals and Objectives	
Educational Goal:	Safely and successfully manage an out of hospital cardiac arrest in the patient where COVID-19 is suspected or cannot be ruled out.
Objectives: (Medical and CRM)	1. Appropriate don and doff airborne PPE in the patient where COVID-19 cannot be ruled out and an aerosol generating medical procedure is performed. 2. Manage cardiac arrest in the patient where COVID-19 cannot be ruled out. 3. Demonstrate effective crisis resource management including role assignment and closed loop communication.
EPAs Assessed:	N/A

Learners, Setting and Personnel			
Target Learners:	☐ Junior Learners	☐ Senior Learners	☒ Staff
	☒ Physicians	☒ Nurses	☒ RTs
	☐ Other Learners:		
Location:	☐ Sim Lab	☒ In Situ	☐ Other:
Recommended Number of Facilitators:	Instructors: 1		
	Confederates: 0		
	Sim Techs: 1		

Scenario Development	
Date of Development:	April 16, 2020
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Revised By:	
Version Number:	



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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Tom Philips		Age: 58		Gender: Male	Weight: 90Kg
Presenting complaint: Post cardiac arrest					
Temp: 38.4C	HR: 110	BP: 80/60	RR: 12 with BVM on iGel SGD	O ₂ Sat: 92%	FiO ₂ : 100%
Cap glucose: 6.2			GCS: 3 (E1 V1 M1)		
Triage note: Patient being transported by advanced care paramedic crew, post VF arrest. Pt was found unresponsive by friend, immediate bystander CPR prior to EHS. EHS arrived and performed CPR. ROSC was obtained and the patient's vitals afterwards were HR 110 sinus tachycardia, BP 80/60, no intrinsic respiratory effort, being ventilated with an iGel supraglottic device in place, GCS 3. ECG at the scene showed anterior ST elevation.					
Allergies: Sulfa drugs					
Past Medical History: HTN COPD 1 pack/day smoker "Frequent Pneumonia"			Current Medications: Ramipril 2.5mg PO BID Salbutamol inhaler prn Tiotropium bromide inhaler		

Section 2B: Extra Patient Information

A. Further History	
Following to be reported by paramedic (info gathered from bystander):	
Patient awoke at 0800, complained of chest pain, SOB, and mild cough. He smoked some marijuana and went back to bed. At 1000 he got up, came into the living room and collapsed. Pt had been complaining of productive cough, fever, moderate SOB and lethargy for 7 days. It is unknown if patient had any contact with COVID-19 positive patients. The patient had not seen their family doctor, felt this was just a normal bout of pneumonia and was taking self-prescribed antibiotics left over from past treatments (clarithromycin).	
The patient was treated with full ACLS modified for COVID-19 including 2 large bore IV's, 1mg EPI Q3 mins x5 doses, 450 mg Amiodarone (300mg +150mg second dose), multiple defibrillations, and ventilated through an iGel supraglottic device. ROSC at 10:25, vitals absent again 2 minutes prior to arrival in the ED.	
B. Physical Exam	
<i>List any pertinent positive and negative findings</i>	
Cardio: Cardiac arrest	Neuro: Unresponsive
Resp: Course crackles bilaterally with ventilation	Head & Neck: Nil
Abdo: Nil	MSK/skin: Nil
Other: Nil	

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Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin (<i>specify type and whether infant/child/adult</i>)
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Mock airborne PPE Defibrillator Intubation equipment
C. Required Medications
Epinephrine Norepinephrine STEMI meds (ASA, ticagrelor, heparin, etc)
D. Moulage
E. Monitors at Case Onset
☒ Patient on monitor with vitals displayed
☐ Patient not yet on monitor
F. Patient Reactions and Exam
<i>Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.</i> Airway – actively being ventilated through iGel supraglottic device Breathing – 100% FiO2 through SGD, bibasilar crackles Circulation – periphery cool to touch and pulses thready

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Section 4: Confederates and Standardized Patients

Confederate and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
None	None required. Use in situ personnel in their usual role as best possible.



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Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: VF HR: 0 BP: 0 RR: 12 via BVM on SGD O ₂ SAT: n/a T: 38.4°C GCS: 3	<i>Is the patient alert? In distress? Seizing? What symptoms do they currently have?</i>	<u>Expected Learner Actions</u> ☐ Safely don airborne PPE ☐ Take handover from EHS ☐ Continue ACLS management ☐ Continue chest compressions with ventilations every 6 seconds ☐ Apply defibrillator pads and conduct rhythm check, shock ☐ Notify cath lab of patient with prehospital STEMI now in cardiac arrest with suspected COVID-19	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> <u>Triggers</u> <i>For progression to next state</i> - Shock x1 → 2. ROSC	
2. ROSC Rhythm: Sinus tachy HR: 110 BP: 80/60 RR: 12 via BVM on SGD O ₂ SAT: 92% T: 38.4°C GCS: 3		<u>Expected Learner Actions</u> ☐ Full set of vitals, ECG, labs, COVID-19 swab, consider XR ☐ Start vasoactive infusion ☐ Liaise with cardiology/cath lab now that ROSC achieved with repeat ECG showing STEMI ☐ Secure definitive airway ☐ Consult ICU	<u>Modifiers</u> Norepinephrine infusion → BP increased to 95/75 <u>Triggers</u> - All actions complete or 5-6 mins → 3. Transfer & handover	
3. Transfer & Handover VS unchanged		<u>Expected Learner Actions</u> ☐ Handover to cath lab team ☐ Patient transported to cath lab with airborne precautions ☐ Update patient's family	<u>Modifiers</u> <u>Triggers</u> - Actions complete → 4. Doff PPE	
4. Doff PPE		<u>Expected Learner Actions</u> ☐ Have trained observer guide through PPE doffing of entire team	End Case	



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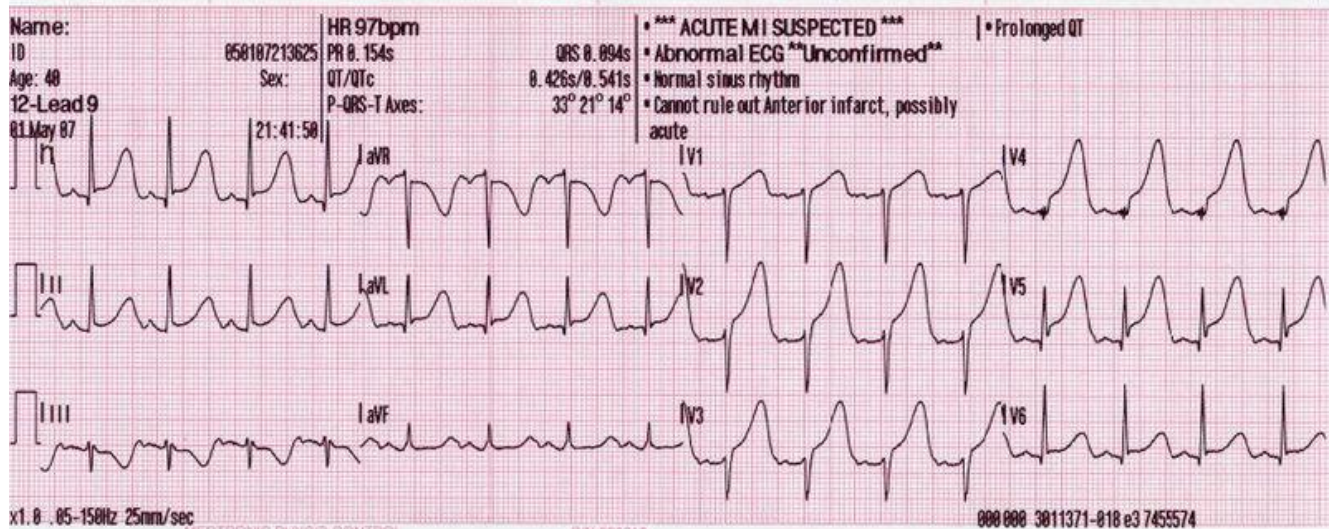
Appendix A: Laboratory Results

No blood work given during case

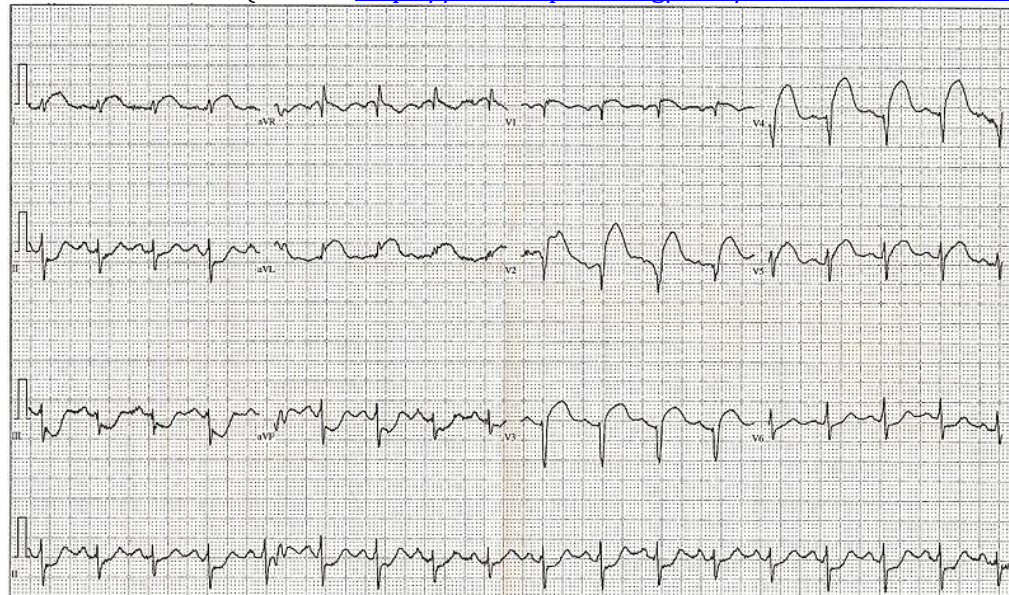
Appendix B: ECGs, X-rays, Ultrasounds and Pictures

Paste in any auxiliary files required for running the session. Don't forget to include their source so you can find them later!

Prehospital ECG (Source: https://commons.wikimedia.org/wiki/File:12_lead_generated_anterior_ML.JPG)



ED ECG Post ROSC (Source: https://en.wikipedia.org/wiki/File:12_Lead_EKG_ST_Elevation_tracing_only.jpg)



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Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

This case was designed to simulate the pathway of a patient going from pre-hospital through to ED with suspicion of COVID-19 and ongoing aerosol generating medical procedure needs.

There are multiple guidelines out there depending on your institution and location. We have intentionally not included any specific guidelines for PPE or for managing cardiac arrest in the suspected COVID-19 patient (aside from the broad AHA guidelines) so that you can build your own institutional guidelines into your simulation and debrief.

Here are some suggested debrief points to discuss:

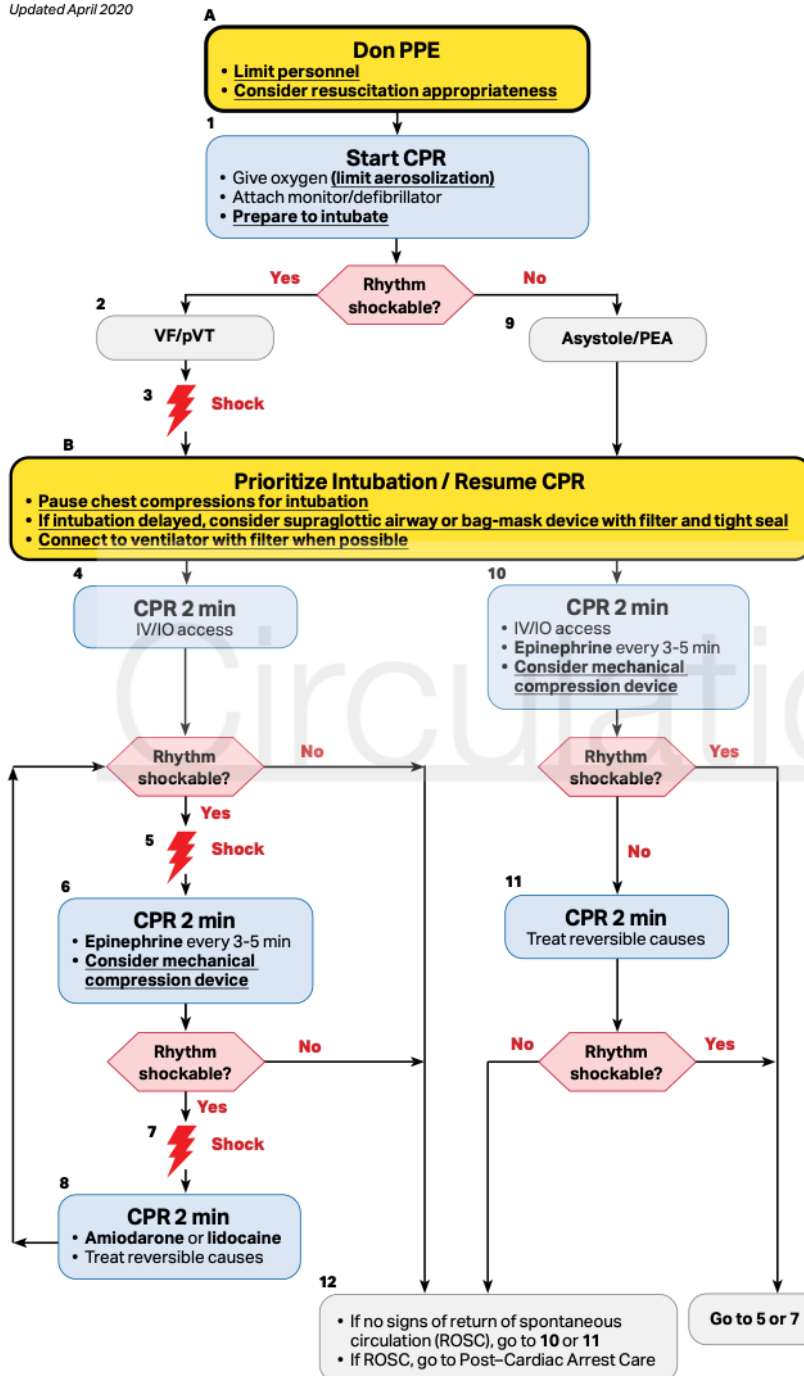
1. What PPE should prehospital providers be wearing in the field when dealing with aerosol generating medical procedures?
2. How best should cardiac arrest and airway compromise be managed in the field?
3. On arrival to the ED with active CPR, where should the transition of care and handover occur when COVID-19 is suspected? In the ambulance bay, in the resuscitation bay, after ROSC?
4. Who needs to be in the room during the management of cardiac arrest? Are there any ways to minimize personnel and therefore minimize exposure?
5. What are some of the major differences between traditional ACLS and ACLS in the suspected COVID-19 patient?
6. What are your institutional guidelines with regard to suspected COVID-19 patients being transferred to the cath lab or OR etc?
7. What are some of the challenges with communication with full airborne PPE? Are there any ways around this?
8. Is there a role for a mechanical compression device in this situation?
9. What if you are in a community setting without a cath lab? What are the implications for transport or thrombolytics?



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ACLS Cardiac Arrest Algorithm for Suspected or Confirmed COVID-19 Patients

Updated April 2020



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CPR Quality

- Push hard (at least 2 inches [5 cm]) and fast (100-120/min) and allow complete chest recoil.
- Minimize interruptions in compressions.
- Avoid excessive ventilation.
- Change compressor every 2 minutes, or sooner if fatigued.
- If no advanced airway, 30:2 compression-ventilation ratio.
- Quantitative waveform capnography
 - If PETCO₂ <10 mm Hg, attempt to improve CPR quality.
- Intra-arterial pressure
 - If relaxation phase (diastolic) pressure <20 mm Hg, attempt to improve CPR quality.

Shock Energy for Defibrillation

- Biphasic:** Manufacturer recommendation (eg, initial dose of 120-200 J); if unknown, use maximum available. Second and subsequent doses should be equivalent, and higher doses may be considered.
- Monophasic:** 360 J

Advanced Airway

- Minimize closed-circuit disconnection
- Use intubator with highest likelihood of first pass success
- Consider video laryngoscopy
- Endotracheal intubation or supraglottic advanced airway
- Waveform capnography or capnometry to confirm and monitor ET tube placement
- Once advanced airway in place, give 1 breath every 6 seconds (10 breaths/min) with continuous chest compressions

Drug Therapy

- Epinephrine IV/IO dose:** 1 mg every 3-5 minutes
- Amiodarone IV/IO dose:** First dose: 300 mg bolus. Second dose: 150 mg.
- or
- Lidocaine IV/IO dose:** First dose: 1-1.5 mg/kg. Second dose: 0.5-0.75 mg/kg.

Return of Spontaneous Circulation (ROSC)

- Pulse and blood pressure
- Abrupt sustained increase in PETCO₂ (typically ≥40 mm Hg)
- Spontaneous arterial pressure waves with intra-arterial monitoring

Reversible Causes

- Hypovolemia
- Hypoxia
- Hydrogen ion (acidosis)
- Hypo-/hyperkalemia
- Hypothermia
- Tension pneumothorax
- Tamponade, cardiac
- Toxins
- Thrombosis, pulmonary
- Thrombosis, coronary

Source: <https://www.ahajournals.org/doi/pdf/10.1161/CIRCULATIONAHA.120.047463>

